

## ROCKY MOUNT WILSON, NC, USA

WMO: 723068

Lat: 35.855N

Lon: 77.893W

Elev: 160

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 93759

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.3       | 23.4 | 2.8                           | 6.4 | 27.3 | 7.3 | 8.0 | 29.3 | 21.3                  | 46.4 | 19.4 | 44.6 | 4.9                   | 310  | 0.408 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.9                   | 94.7            | 76.5 | 92.4 | 75.9 | 90.3 | 75.1 | 79.3                | 89.5 | 78.1 | 87.9 | 77.2 | 86.4 | 7.8                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.5                            | 139.1 | 83.2 | 75.4 | 133.7 | 81.8 | 74.6 | 130.2 | 81.0 | 42.7          | 89.3 | 41.5 | 88.3 | 40.5 | 86.6 | 83.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |     |            |     |            |     |            |     |       |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|-----|------------|-----|------------|-----|------------|-----|-------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |     | n=10 years |     | n=20 years |     | n=50 years |     |       |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min | Max        | Min | Max        | Min | Max        |     |       |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |       |
| 18.7              | 16.6 | 14.5 |                            | DB   | 12.7 | 99.1               | 4.7                                                | 3.0       | 9.4 | 101.2      | 6.6 | 103.0      | 4.0 | 104.7      | 0.5 | 106.8 |
|                   |      |      |                            | WB   | 11.0 | 81.1               | 4.3                                                | 1.3       | 8.0 | 82.1       | 5.5 | 82.9       | 3.1 | 83.6       | 0.0 | 84.6  |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 61.0   | 41.1  | 44.3  | 50.9  | 60.9  | 69.0  | 76.7  | 79.6  | 78.3  | 72.7  | 61.8  | 51.2  | 44.6 |      |
|                                                                               | DBStd                                                                         | 15.70  | 10.65 | 9.44  | 9.52  | 8.45  | 7.02  | 4.90  | 3.98  | 4.11  | 5.40  | 8.06  | 8.50  | 9.48 |      |
|                                                                               | HDD50                                                                         | 962    | 316   | 210   | 108   | 11    | 0     | 0     | 0     | 0     | 0     | 7     | 86    | 224  |      |
|                                                                               | HDD65                                                                         | 3211   | 742   | 583   | 444   | 176   | 45    | 1     | 0     | 0     | 8     | 157   | 421   | 634  |      |
|                                                                               | CDD50                                                                         | 4986   | 40    | 50    | 138   | 340   | 588   | 801   | 917   | 877   | 682   | 374   | 121   | 58   |      |
|                                                                               | CDD65                                                                         | 1759   | 1     | 3     | 9     | 54    | 168   | 352   | 452   | 412   | 240   | 59    | 6     | 3    |      |
|                                                                               | CDH74                                                                         | 16033  | 2     | 14    | 87    | 528   | 1526  | 3394  | 4441  | 3750  | 1808  | 444   | 33    | 6    |      |
|                                                                               | CDH80                                                                         | 6523   | 0     | 0     | 9     | 124   | 516   | 1477  | 2043  | 1636  | 618   | 99    | 1     | 0    |      |
| Wind                                                                          | WSAvg                                                                         | 5.8    | 6.4   | 6.8   | 7.1   | 7.1   | 5.9   | 5.3   | 4.9   | 4.3   | 5.2   | 5.3   | 5.5   | 5.6  |      |
| Precipitation                                                                 | PrecAvg                                                                       | 45.6   | 3.6   | 3.2   | 4.1   | 2.9   | 3.6   | 4.0   | 4.8   | 4.7   | 4.9   | 3.2   | 3.0   | 3.3  |      |
|                                                                               | PrecMax                                                                       | 62.9   | 7.2   | 7.6   | 7.2   | 5.6   | 6.8   | 9.8   | 7.7   | 10.2  | 27.2  | 6.8   | 6.7   | 6.1  |      |
|                                                                               | PrecMin                                                                       | 31.4   | 0.9   | 0.6   | 1.3   | 1.0   | 1.3   | 1.9   | 2.5   | 1.0   | 0.3   | 0.0   | 0.5   | 1.0  |      |
|                                                                               | PrecStd                                                                       | 6.8    | 1.6   | 1.5   | 1.6   | 1.2   | 1.4   | 2.3   | 1.6   | 2.4   | 4.8   | 1.7   | 1.8   | 1.3  |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 72.2  | 76.3  | 81.4  | 87.6  | 91.9  | 97.5  | 98.3  | 97.4  | 92.4  | 87.1  | 78.7  | 74.1 |      |
|                                                                               |                                                                               | MCWB   | 63.8  | 64.1  | 63.3  | 69.1  | 72.4  | 76.1  | 77.9  | 76.5  | 73.9  | 73.9  | 67.5  | 66.4 |      |
|                                                                               | 2%                                                                            | DB     | 67.4  | 70.6  | 76.1  | 83.5  | 88.4  | 93.5  | 94.7  | 93.3  | 89.4  | 82.5  | 73.6  | 69.9 |      |
|                                                                               |                                                                               | MCWB   | 61.4  | 60.1  | 61.9  | 66.5  | 71.3  | 75.1  | 77.6  | 76.6  | 73.9  | 69.9  | 62.9  | 63.8 |      |
|                                                                               | 5%                                                                            | DB     | 63.3  | 65.9  | 72.2  | 80.6  | 85.7  | 90.8  | 92.5  | 90.9  | 86.5  | 79.1  | 70.3  | 65.6 |      |
|                                                                               |                                                                               | MCWB   | 57.3  | 57.8  | 59.7  | 65.5  | 70.4  | 74.5  | 76.7  | 76.0  | 72.6  | 68.4  | 63.0  | 61.1 |      |
|                                                                               | 10%                                                                           | DB     | 58.5  | 61.9  | 67.6  | 76.5  | 82.4  | 88.2  | 90.1  | 88.5  | 83.6  | 75.4  | 66.2  | 61.3 |      |
|                                                                               |                                                                               | MCWB   | 52.7  | 54.4  | 57.6  | 63.6  | 69.3  | 73.4  | 76.1  | 75.0  | 71.7  | 67.2  | 59.8  | 56.7 |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 65.9  | 66.3  | 67.0  | 72.9  | 75.6  | 78.6  | 81.3  | 80.6  | 78.3  | 76.4  | 70.4 | 69.0 |
|                                                                               |                                                                               |        | MCDB  | 69.4  | 73.0  | 75.3  | 80.9  | 85.3  | 89.4  | 92.7  | 90.2  | 86.5  | 81.2  | 72.8 | 71.6 |
| 2%                                                                            |                                                                               | WB     | 62.3  | 63.5  | 64.9  | 69.9  | 73.8  | 77.3  | 79.5  | 79.1  | 76.8  | 73.9  | 67.0  | 65.1 |      |
|                                                                               |                                                                               | MCDB   | 66.5  | 68.0  | 72.0  | 78.9  | 83.5  | 88.3  | 90.5  | 88.9  | 84.4  | 78.9  | 71.4  | 68.0 |      |
| 5%                                                                            |                                                                               | WB     | 58.4  | 59.7  | 62.9  | 67.7  | 72.5  | 76.3  | 78.3  | 78.0  | 75.4  | 71.6  | 64.3  | 61.9 |      |
|                                                                               |                                                                               | MCDB   | 62.2  | 65.4  | 68.6  | 76.4  | 81.6  | 86.9  | 88.7  | 87.1  | 81.9  | 76.1  | 68.1  | 65.1 |      |
| 10%                                                                           |                                                                               | WB     | 53.8  | 54.7  | 60.2  | 65.6  | 71.0  | 75.2  | 77.3  | 76.9  | 74.2  | 69.0  | 61.4  | 57.2 |      |
|                                                                               |                                                                               | MCDB   | 57.1  | 60.3  | 66.5  | 73.4  | 79.3  | 85.2  | 87.1  | 85.1  | 79.8  | 73.8  | 66.0  | 60.7 |      |
| Mean Daily<br>Temperature<br>Range                                            |                                                                               | 5% DB  | MDBR  | 19.5  | 21.0  | 22.5  | 23.4  | 21.5  | 20.6  | 19.9  | 19.3  | 19.2  | 21.1  | 22.3 | 19.8 |
|                                                                               |                                                                               |        | MCDBR | 24.7  | 25.6  | 26.7  | 26.5  | 24.4  | 23.1  | 22.5  | 22.3  | 22.8  | 23.8  | 25.1 | 22.6 |
|                                                                               | MCWBR                                                                         |        | 18.2  | 17.9  | 14.7  | 11.9  | 9.8   | 8.1   | 7.7   | 7.8   | 9.1   | 12.0  | 15.6  | 17.1 |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 22.4  | 24.1  | 22.4  | 21.8  | 20.6  | 20.1  | 20.2  | 19.1  | 17.3  | 17.9  | 19.5  | 20.4 |      |
|                                                                               |                                                                               | MCWBR  | 18.6  | 19.2  | 15.1  | 11.1  | 9.2   | 7.9   | 7.9   | 7.9   | 7.8   | 10.1  | 14.1  | 17.3 |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.312  | 0.329 | 0.357 | 0.406 | 0.455 | 0.504 | 0.535 | 0.529 | 0.417 | 0.367 | 0.328 | 0.312 |      |      |
|                                                                               | taud                                                                          | 2.500  | 2.449 | 2.404 | 2.269 | 2.166 | 2.075 | 2.004 | 2.010 | 2.324 | 2.443 | 2.506 | 2.535 |      |      |
|                                                                               | Ebn at Noon                                                                   | 282    | 289   | 289   | 279   | 265   | 252   | 242   | 241   | 266   | 272   | 273   | 275   |      |      |
|                                                                               | Edh at Noon                                                                   | 26     | 31    | 35    | 42    | 48    | 52    | 56    | 54    | 38    | 31    | 26    | 24    |      |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 808    | 1029  | 1354  | 1715  | 1897  | 2027  | 1927  | 1739  | 1438  | 1164  | 890   | 709   |      |      |
|                                                                               | RadStd                                                                        | 56     | 116   | 96    | 122   | 144   | 128   | 106   | 95    | 115   | 148   | 71    | 58    |      |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | +1.08 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +342  | +265  |
| (47) | Regional (65 neighbors) | +0.79 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +277  | +191  |

Nomenclature: See separate page